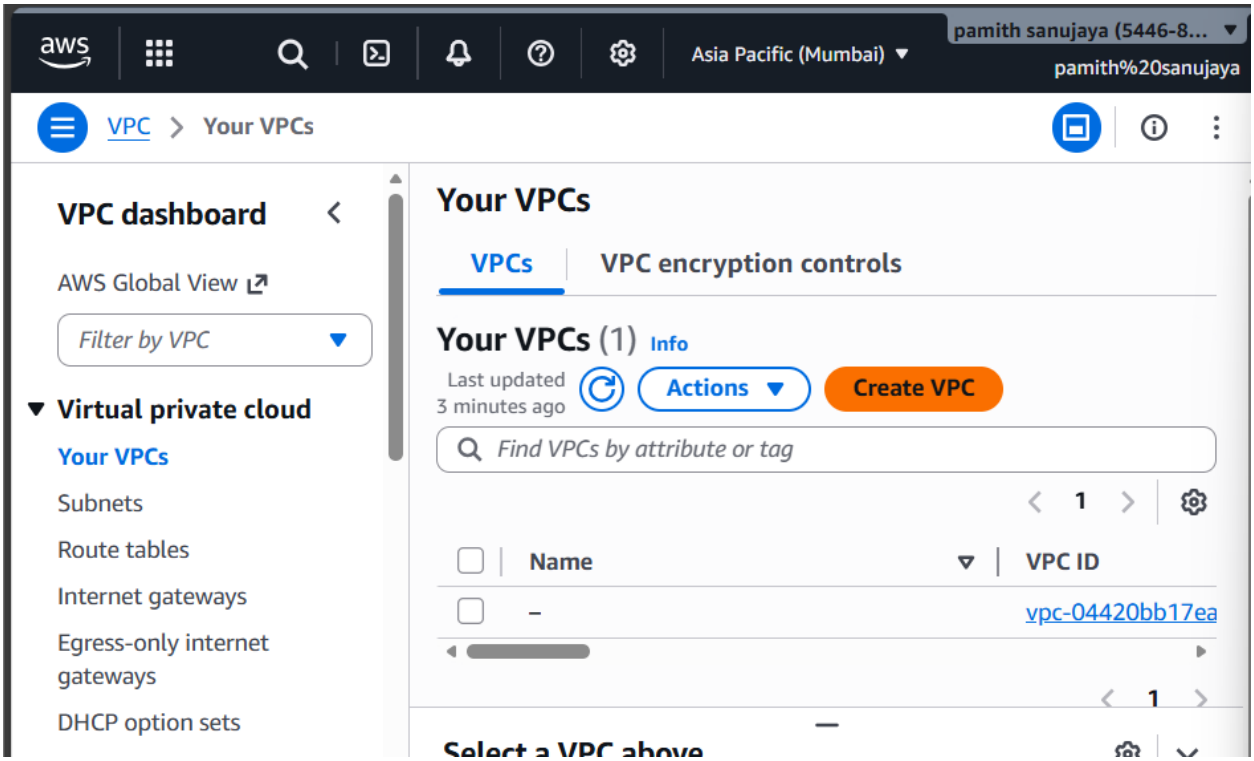
How to create a VPC in AWS

This project demonstrates the fundamental principles of cloud infrastructure security and network segmentation through the deployment of a Virtual Private Cloud (VPC) within Amazon Web Services (AWS).

This project serves as a practical exercise to showcase knowledge in AWS networking. The objective is to establish a robust and isolated network environment, laying the groundwork for deploying secure applications and services in the cloud.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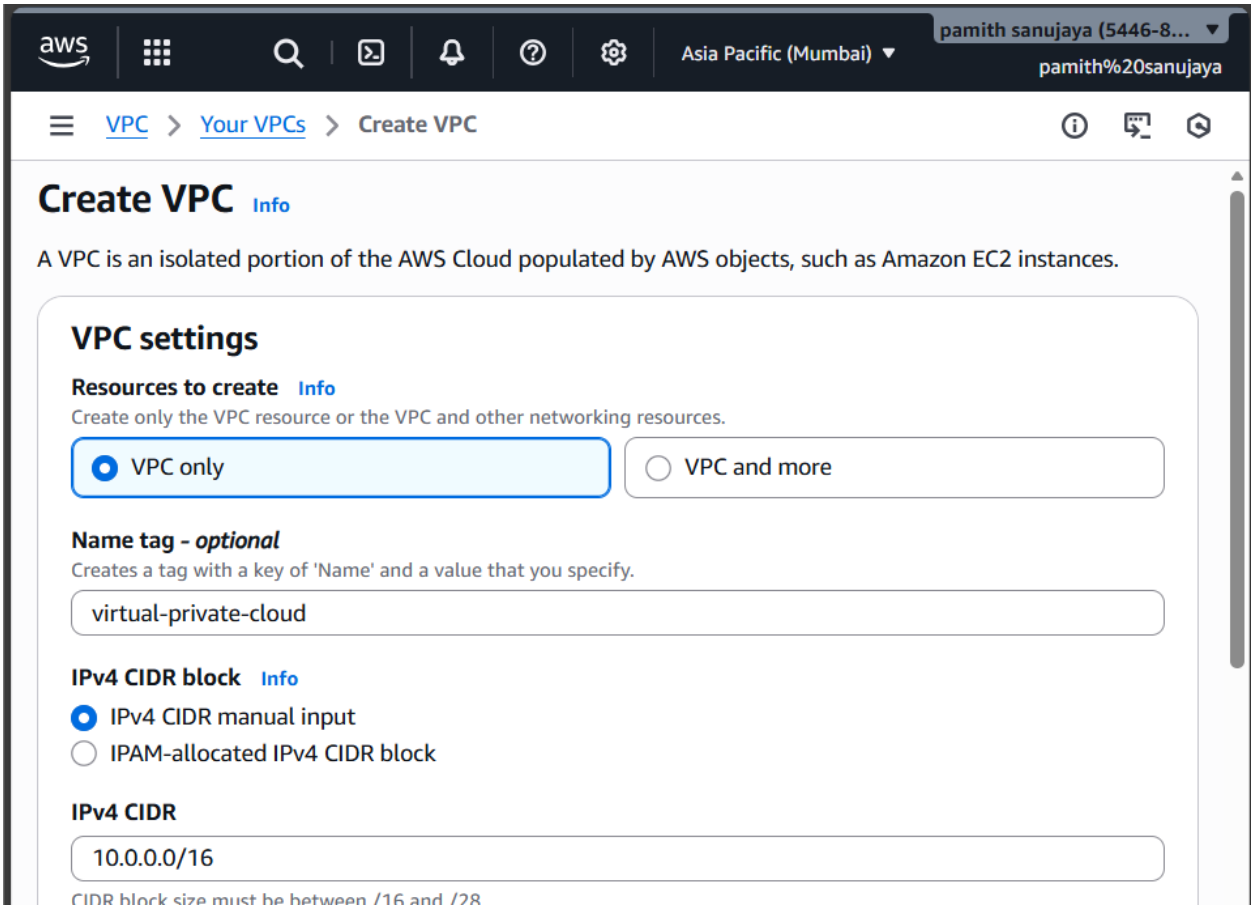
VPC > Your VPCs
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Your VPCs
VPCs
VPC encryption controls
Your VPCs (1) Info
Last updated 3 minutes ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
–
vpc-04420bb17ea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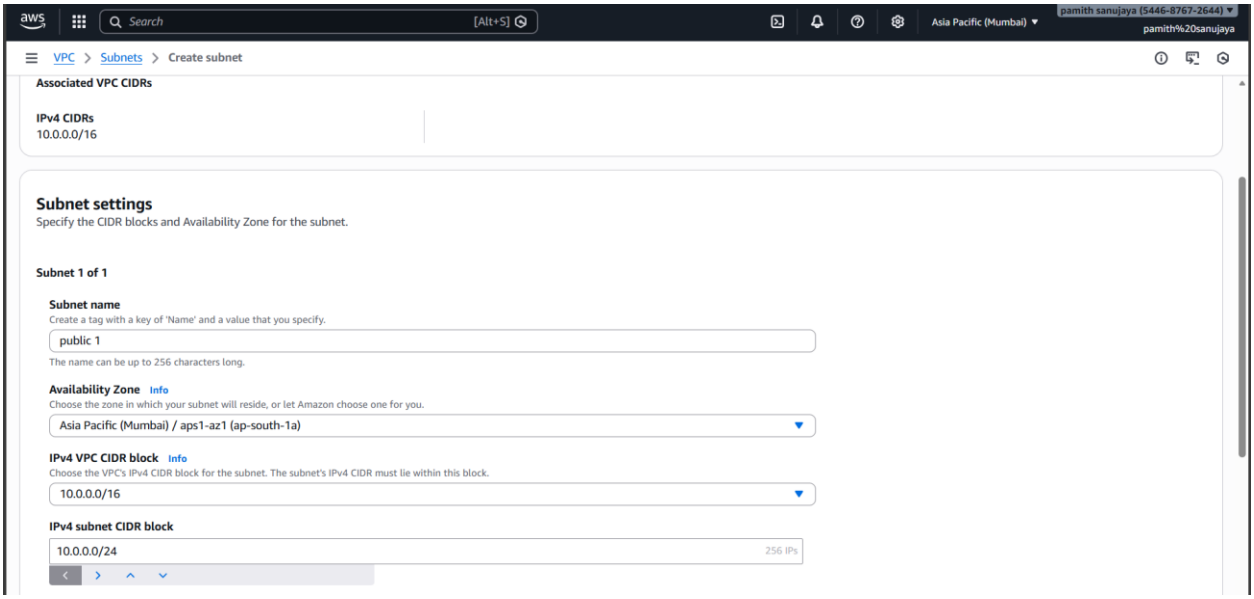
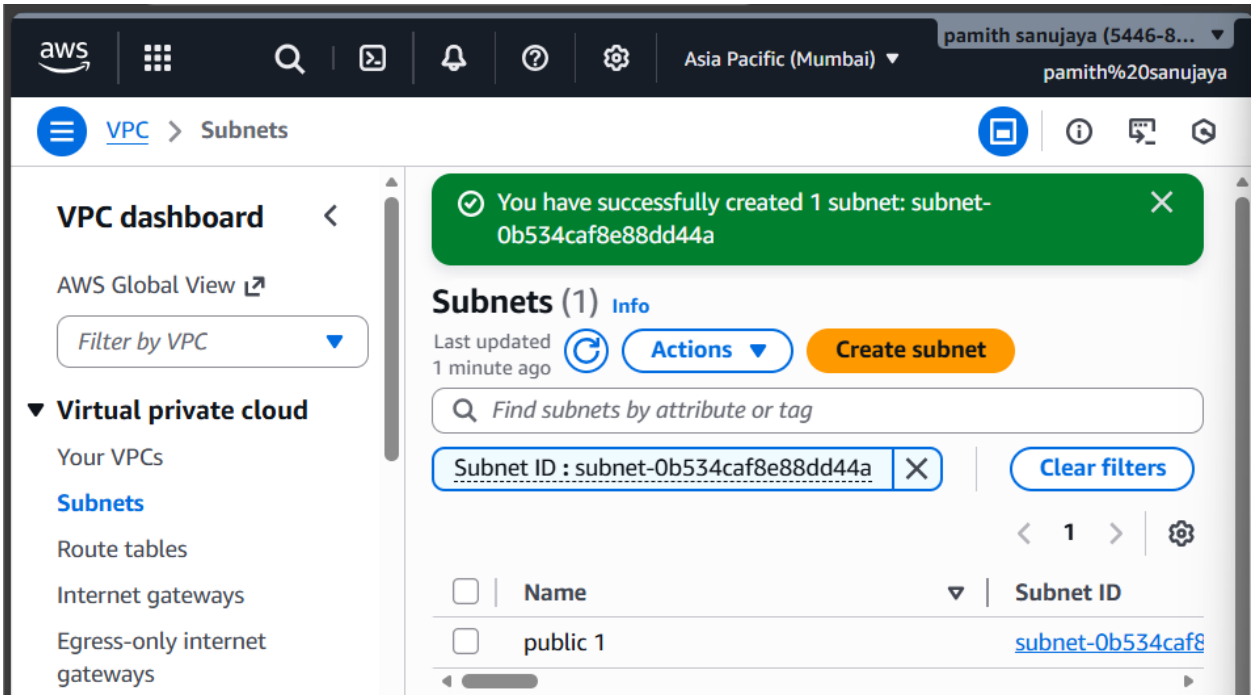
VPC > Your VPCs > Create VPC
Create VPC Info
A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.
VPC settings
Resources to create Info
Create only the VPC resource or the VPC and other networking resources.
VPC only
VPC and more
Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.
virtual-private-cloud
IPv4 CIDR block Info
IPv4 CIDR manual input
IPAM-allocated IPv4 CIDR block
IPv4 CIDR
10.0.0.0/16





Enable auto assign ip address and to make accessible to public internet

VPC > Subnets > subnet-0b534caf8e88dd44a > Edit subnet settings

Edit subnet settings Info

Subnet

Subnet ID
subnet-0b534caf8e88dd44a

Name
public 1

Auto-assign IP settings Info
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☐ Enable auto-assign public IPv4 address Info

☐ Enable auto-assign customer-owned IPv4 address Info
Option disabled because no customer owned pools found.

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VPC > Internet gateways > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

internet gateway

Tags - *optional*

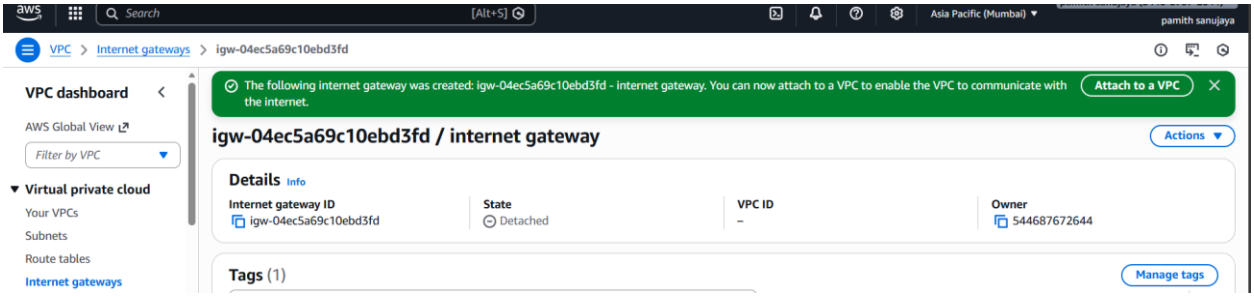
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - *optional*	
Name	internet gateway	Remove

Add new tag

You can add 49 more tags.

Cancel | Create internet gateway


aws
Search
[Alt+S]
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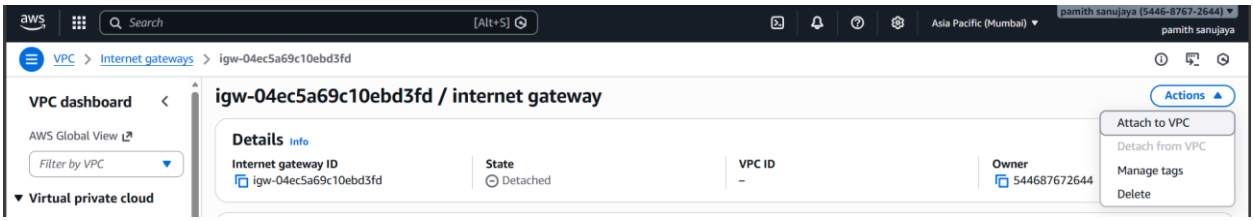
VPC > Internet gateways > igw-04ec5a69c10ebd3fd
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
The following internet gateway was created: igw-04ec5a69c10ebd3fd - internet gateway. You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC
igw-04ec5a69c10ebd3fd / internet gateway
Actions
Details Info
Internet gateway ID
igw-04ec5a69c10ebd3fd
State
Detached
VPC ID
–
Owner
544687672644
Tags (1)
Manage tags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Search
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Detach from VPC
Manage tags
Delete
Details Info
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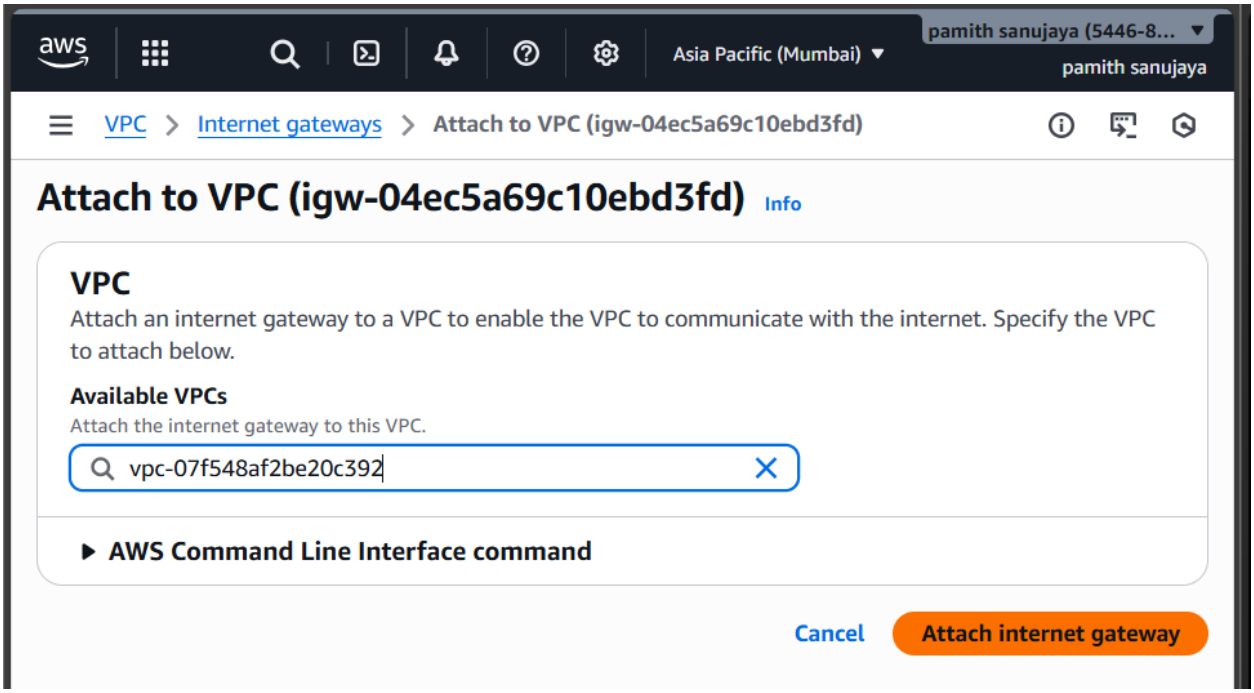
igw-04ec5a69c10ebd3fd
State
Detached
VPC ID
–
Owner
544687672644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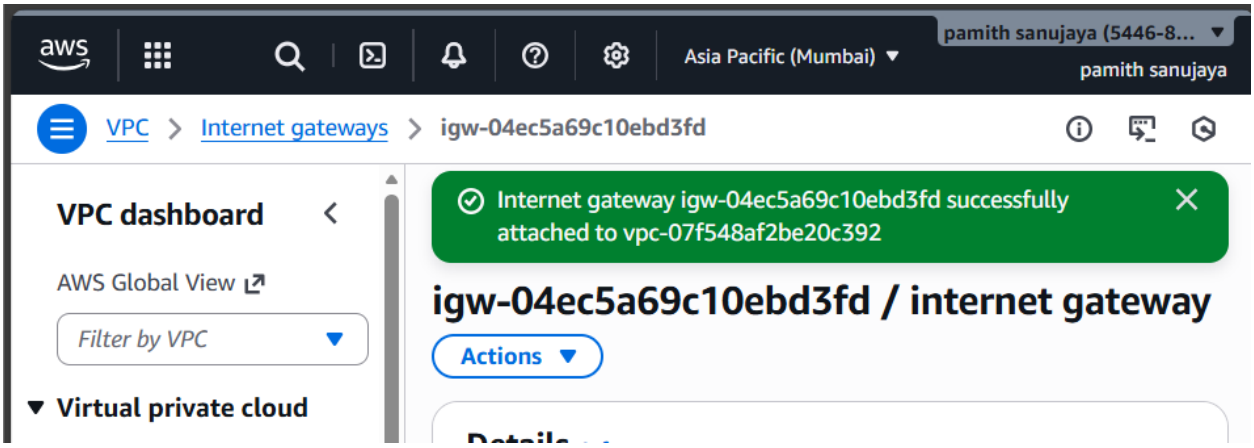
VPC > Internet gateways > Attach to VPC (igw-04ec5a69c10ebd3fd)
Attach to VPC (igw-04ec5a69c10ebd3fd) Info
VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the internet gateway to this VPC.
vpc-07f548af2be20c392
AWS Command Line Interface command
Cancel
Attach internet gateway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VPC > Internet gateways > igw-04ec5a69c10ebd3fd
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Internet gateway igw-04ec5a69c10ebd3fd successfully attached to vpc-07f548af2be20c392
igw-04ec5a69c10ebd3fd / internet gateway
Actions